7. Technology Stack

Date	26 June 2026
Team ID	SWTID-2026-8135
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, Jinja2 templates, vanilla CSS (glassmorphic design), vanilla JavaScript	No build step required; templates render server-side from FastAPI, keeping the stack lightweight for a fast demo turnaround.
2	Backend / Server-Side	Python 3.10, FastAPI, Uvicorn	Async-first framework with automatic OpenAPI docs, native Pydantic validation, and fast iteration for REST endpoints.
3	Database / Data Storage	JSON flat files (data/*.json) via utils/db_manager.py	Zero-setup persistence that mirrors a relational schema (Users, Queries, Responses, Quizzes, Summaries, Learning Paths) without requiring an external DBMS for a demo/academic project.
4	Cloud / Hosting / Deployment	Docker (python:3.10-slim base image), Uvicorn ASGI server	Containerization ensures the app runs consistently across development and evaluation environments; exposed on port 8000.
5	Version Control & CI/CD	Git, GitHub	Standard collaborative source control for the team; no automated CI pipeline is configured yet.
6	Third-Party APIs / Other Tools	Google Gemini API (google-generativeai), HuggingFace Transformers + PyTorch running MBZUAI/LaMini-Flan-T5 locally	Combines cloud-grade reasoning quality (Gemini, with dynamic best-model resolution) with an offline-capable local model for resilience when no API key or internet access is available.